PAPER_284: M16 Eagle Nebula UQFF — Dual Mass Co-Action Product (Φ_dm)

SFR Growth × Photoevaporation Erosion Multiplicative Coupling

Classification: UQFF 2.0 Gravitational Physics — Nebular Mass Dynamics
System: M16 Eagle Nebula (IC 4703), Eagle Nebula Star-Forming Region
Session: 80 | **Module:** M16_UQFF_MODULE.cpp (22nd C++ UQFF module)
Author: Daniel T. Murphy | **Date:** March 2026

Abstract

This paper introduces the **Dual Mass Co-Action Product** (Φ_dm) — a UQFF gravity modulation factor that couples star-formation-driven mass accumulation and radiation-driven photoevaporation erosion through a **multiplicative** product rather than the previously used additive form. For M16 (Eagle Nebula), with star formation rate SFR = 1 $M_\odot$/yr over initial gas mass M_0 = 1200 $M_\odot$, and maximum photoevaporation fraction E_0 = 0.3 (30%), the multiplicative form produces a 24.3% reduction in Φ_dm relative to the additive approximation at t = 5 Myr. This is the **first UQFF module** to simultaneously apply an additive-gain and saturation-subtractive product on the same gravity term.

---

2. Physical Motivation

In active star-forming nebulae, two competing processes drive mass evolution:

1. **Star Formation Accretion** — molecular gas accretes onto protostars, increasing the effective gravitational mass fraction by SFR_rate × t.
2. **Photoevaporation Erosion** — UV radiation from newly formed massive stars erodes the surrounding gas, progressively reducing the effective mass by a saturating fraction E_rad(t).

The **additive approximation** (used in prior UQFF modules):

$$\Phi_{dm}^{add} = g_{base} \times (1 + M_{sf}) - E_{rad}$$

linearly superposes the two effects, implicitly treating them as independent processes acting on separate mass reservoirs.

The **multiplicative form** (this paper):

$$\Phi_{dm}(t) = (1 + \text{SFR\_rate} \times t) \times (1 - E_{rad}(t))$$

correctly encodes that **the eroded mass is drawn from the same growing reservoir** — the fraction lost to photoevaporation scales with the mass being accreted, not the original quiescent mass. This is physically accurate for pillar-geometry star formation (e.g., M16's "Pillars of Creation").

---

3. Mathematical Formulation

3.1 Parameters (M16 Eagle Nebula)

Parameter	Value	Description
M_0	2.387×10^{33} kg (1200 M⊙)	Initial nebula gas mass
SFR	1 M⊙/yr	Star formation rate
SFR_rate	2.639×10^{-11} s^{-1}	= SFR / (M_0/M⊙) / (3.156×10^7 s/yr)
τ_erode	9.468×10^{13} s (3 Myr)	Photoevaporation e-folding time
E_0	0.3	Maximum photoevaporation fraction
g_base	1.454×10^{-12} m/s^2	G × M / r^2 at r = 3.31×10^{17} m

3.2 Dual Co-Action Product

$$M_{sf}(t) = \text{SFR\_rate} \times t$$

$$E_{rad}(t) = E_0 \left(1 - e^{-t/\tau}\right)$$

$$\boxed{\Phi_{dm}(t) = (1 + M_{sf}) \times (1 - E_{rad})}$$

3.3 Dynamic Gravity Term

$$g_{dyn}(t) = g_{base} \times \Phi_{dm}(t)$$

3.4 Multiplicative–Additive Gap

The gap relative to the additive form:

$$\Delta_{gap} = \Phi_{dm}^{mult} - \Phi_{dm}^{add} = -(M_{sf} \times E_{rad})$$

This cross-term is always **negative** — the multiplicative form predicts lower gravity than the additive approximation whenever both SFR accumulation and erosion are simultaneously active.

---

4. Numerical Results at t = 5 Myr

t = 5 Myr = 1.578×10^{14} s

Quantity	Value
M_sf_frac	SFR_rate × t = 2.639×10^{-11} × 1.578×10^{14} = **4164.8**
E_rad	E_0 × (1 − exp(−5/3)) = 0.3 × 0.8110 = **0.2433**
Φ_dm (multiplicative)	(1 + 4164.8) × (1 − 0.2433) = 4165.8 × 0.7567 = **3151.9**
Φ_dm (additive)	(1 + 4164.8) − 0.2433 = **4165.6**
gap_mult_add	−(4164.8 × 0.2433) = −**1013.3** (24.3% less)
g_dyn(5 Myr)	1.454×10^{-12} × 3151.9 = **4.583 × 10^{-9} m/s^2**

The multiplicative gap of -1013.3 confirms that treating erosion as acting on the growing mass reservoir (not the static initial mass) produces a **measurable 24.3% reduction** compared to the additive approximation.

---

5. Connection to UQFF 2.0 g_total

In the full M16 UQFF 2.0 equation:

$$g_{total}(r,t) = \left[g_{dyn}(t) + U_{g,sum}(26) + \Lambda + Q + L + F + g_{exp}\right] \times \mathrm{corr}_{SC}$$

The Φ_dm product modulates only the dynamic base gravity term g_dyn. The 26-layer Triadic (U_g,sum), cosmological Λ, quantum, Lorentz, fluid, and Friedmann expansion terms are all independent of Φ_dm — the modulation is cleanly scoped to the time-evolving mass component.

---

6. UQFF Historical Distinction

Module	SFR Term	Erosion Term	Form
Session 55 CP3 M16EagleNebulaRadiationSFR	g_base×(1+M_sf)	−E_rad	**Additive**
This paper (PAPER_284)	(1+M_sf)	×(1−E_rad)	**Multiplicative**

This is the **first UQFF module** to use the multiplicative dual co-action form, correctly encoding the coupled feedback between star formation accretion and pillar photoevaporation for M16-class nebulae.

---

7. Wolfram KB Term

```
M16UQFF:Phi_dm=(1+SFR_rate*t)*(1-E_0*(1-Exp[-t/tau]));    SFR_rate=2.639e-11/s; M_sf_frac=SFR_rate*t [PAPER_284]
```

---

8. Cross-References

- **PAPER_285:** Erosion Saturation Half-Time (t_half, ΔgMax)
- **PAPER_286:** Nebular Friedmann Redshift (κ_neb, z=0.0015)
- **M16_UQFF_MODULE.cpp:** Full UQFF 2.0 C++ implementation (22nd module)
- **CondensedPhysics3.py:** `M16DualMassCoActionProductCalculator`

---

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

---

§B. VDS/DVP/BSH Deep Synthesis

§B.1 Vacuum Density Series (VDS)

The canonical VDS ratio $\rho_{\mathrm{vac},[\mathrm{SCm}]}/\rho_{\mathrm{UA}} = 1.894$ governs the double-exponential vacuum condensate profile: $\rho_{\mathrm{vac}}(r) = \rho_{\mathrm{vac},[\mathrm{SCm}]} \cdot \exp\left(-\exp\left(-\frac{r-r_0}{\lambda_{\mathrm{VDS}}}\right)\right)$

---

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

For this system, the local VDS sub-ratio is 0.126 (near-threshold regime), placing it in the $t \to \pi$ collapse zone where the double-exponential transitions sharply from condensed to dilute vacuum. This threshold behavior connects

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§B.2 Dipole Vortex Primes (DVP)

The DVP encoding maps the system's characteristic parameter onto the prime lattice: $p_{\text{DVP}} = 47, \quad n_{\text{channel}} = 25/26$

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

Since $p_{\mathrm{DVP}} = 47$ is **resonant** (threshold at $p > 26$), the system's vacuum topology inherits its resonant enhancement from the DVP lattice, amplifying UQFF coupling at specific radii where compressed matter

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

---

§B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is **10^6 yr** (Jeans collapse timescale):

$$\mathcal{F}_{\text{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_{U_b} \cdot \left(1 - e^{-[SSq]\cdot m/M_\odot}\right) \cdot \cos\left(\frac{2\pi j}{26}\right)$$

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

The tanh saturation envelope prevents unphysical divergence: $\mathcal{F}_{\text{BSH,sat}} = \mathcal{F}_{\text{BSH}} \cdot \left(1 - \tanh\left(\frac{t-t_{\text{sat}}}{\tau_{\text{BSH}}}\right)\right)$

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

connecting to the PAPER_877 Stage 5 buoyancy seed $U_{b,\text{seed}} = 0.1 \cdot (\hbar c/r^2) \cdot f_{\text{SCm}}$ which initializes the harmonic series at cosmogenesis.

§B.4 Production-Scale Consistency

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

Framework	Canonical Value	This Paper	Status
VDS ratio	$\rho_{\mathrm{SCm}}/\rho_{\mathrm{UA}} = 1.894$	Local sub-ratio = 0.126	✓ Threshold-consistent
DVP prime	$p_k \in$ {2,3,...,113}	$p_{\mathrm{DVP}} = 47$	✓ Res-

§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Fine structure constant α	UQFF reproduces α via Ug1 dipole coupling	1/137.036	PDG 2024	✓ Consistent
Cosmological constant Λ	1.1×10^{-52} m^{-2} (UQFF vacuum term)	1.114×10^{-52} m^{-2}	Planck 2018	✓ Consistent
Proton decay rate	κ = 0.0005/day $\rightarrow$ Γ_p suppression	< 4.17×10^{-35}/yr	Super-K 2024	✓ Consistent
UQFF buoyancy signature	F_U_Bi_i unique gravitational correction	Not yet measured	Future gravitational wave detectors	Testable

New physics claim: UQFF introduces buoyancy-based gravitational corrections (F_U_Bi_i) that produce measurable deviations from GR at scales where vacuum condensate density ρ_SCm becomes significant, offering a falsifiable prediction beyond the Standard Model.

Cross-validated with PAPER_642 (`UQFFSMParameterBridgeMasterComparisonCalculator`) for full UQFF–SM bridge.